

Theories — Good and Bad

Chris Garratt (U. Victoria):

Q. Do you want to hear my theory of the Gulf Stream?

A. It's the Navier-Stokes equations.

So our theory of climate is

- (i) The Navier-Stokes equations
- (ii) Thermodynamical laws
- (iii) Radiative transfer (EM) equations
- (iv) Boundary conditions

This is what a GCM is — so GCMs are our theory? Yes, in some sense, but it is

- ❶ low level, and so not very economical;
- ❷ in parts ad hoc, and so not very general.

and therefore, perhaps, not a good theory. Is there a better, 'higher level' theory?